PROBLEM6:

PROBLEM STATEMENT:

Multi-Plan Telecommunications Cost Optimization: Evaluate several mobile plans against usage, validity, and budget constraints; calculate excess charges and identify the most economical eligible plan

OBJECTIVE:

To develop a C program that evaluates multiple mobile plans based on data usage, call minutes, validity, and budget constraints. The program calculates the total cost, including excess usage charges, and identifies the most economical eligible mobile plan.

PROGRAM:

```
#include <stdio.h>


int main() {
        int n, i, best = -1;
        float budget, dataUsed, extraCost;
        float minCost = 999999, totalCost;

        printf("Enter number of mobile plans: ");
    scanf("%d", &n);

        printf("Enter your data usage (GB): ");
    scanf("%f", &dataUsed);

        printf("Enter your budget: ");
    scanf("%f", &budget);

        for (i = 1; i <= n; i++) {
        float price, dataLimit, extraRate;
```

```
    int validity, requiredDays;

printf("\nEnter details for Plan %d\n", i);
printf("Plan price: ");
scanf("%f", &price);

printf("Data limit (GB): ");
scanf("%f", &dataLimit);

printf("Validity (days): ");
scanf("%d", &validity);

    printf("Required validity (days): ");
scanf("%d", &requiredDays);

printf("Extra charge per GB: ");
scanf("%f", &extraRate);

    extraCost = 0;

    if (dataUsed > dataLimit)
    extraCost = (dataUsed - dataLimit) * extraRate;

    totalCost = price + extraCost;

printf("Total cost: Rs. %.2f\n", totalCost);

    if (validity >= requiredDays &&
    totalCost <= budget &&
    totalCost < minCost) {
    minCost = totalCost;
```

```
            best = i;
            }
            }

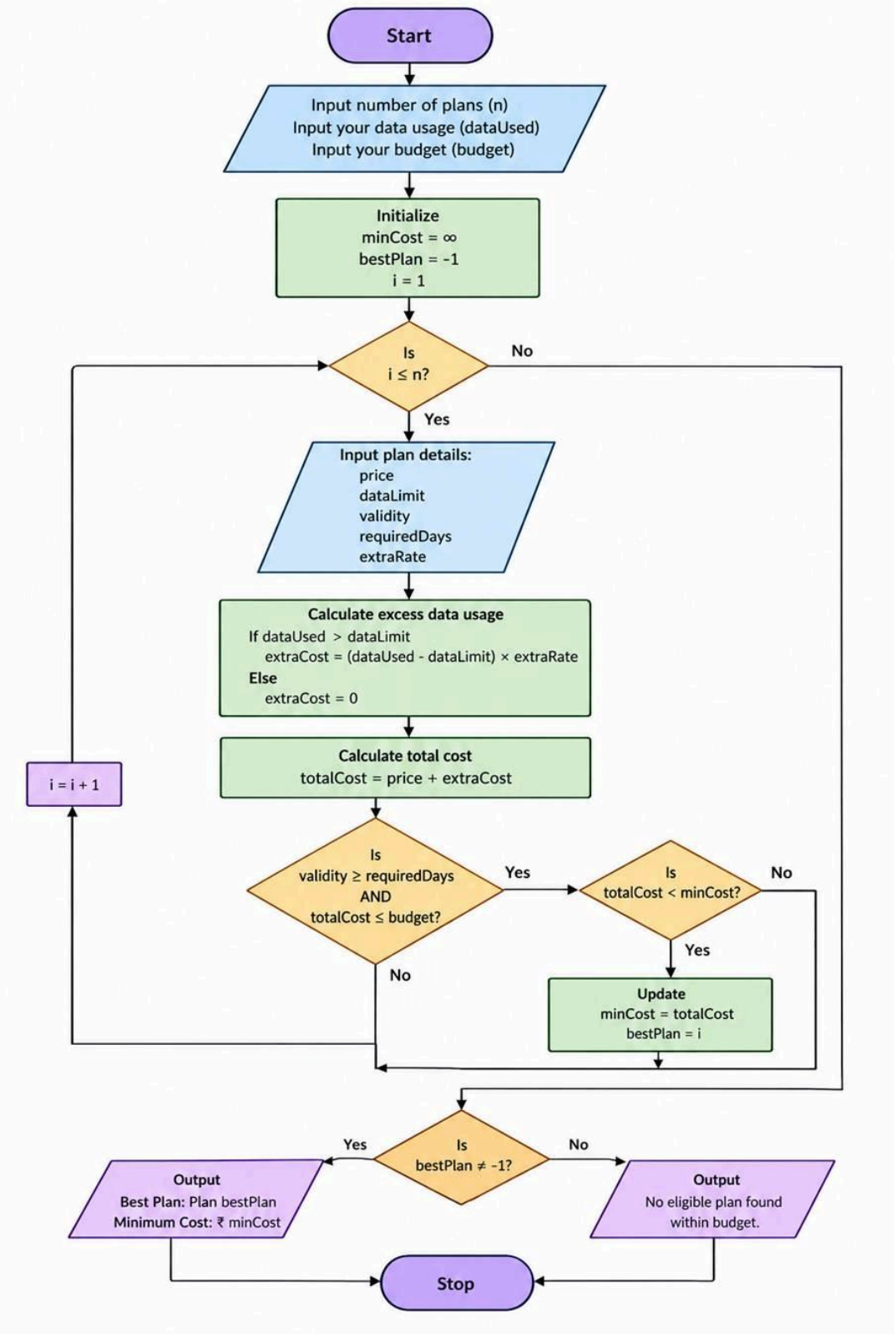
            if (best != -1) {
        printf("\nBest Plan: Plan %d\n", best);
        printf("Minimum Cost: Rs. %.2f\n", minCost);
            } else {
        printf("\nNo eligible plan found within budget.\n");
            }

            return 0;
}
```

ALGORITHM:

1. Start.
2. Read the number of mobile plans.
3. Read the user's data usage and budget.
4. For each mobile plan:
    - Read plan price, data limit, validity, required validity, and extra charge per GB.
    - Calculate excess data usage, if any.
    - Calculate the total cost.
    - Check whether the plan satisfies the validity and budget constraints.
    - Update the minimum cost and best plan if the eligible plan is cheaper.
5. If an eligible plan exists, display the best plan and its cost.
6. Otherwise, display “No eligible plan found within budget.”
7. Stop.

FLOWCHART:


Start
Input number of plans (n)
Input your data usage (dataUsed)
Input your budget (budget)
Initialize
minCost = ∞
bestPlan = -1
i = 1
Is
i ≤ n?
No
Yes
Input plan details:
price
dataLimit
validity
requiredDays
extraRate
Calculate excess data usage
If dataUsed > dataLimit
extraCost = (dataUsed - dataLimit) × extraRate
Else
extraCost = 0
Calculate total cost
totalCost = price + extraCost
i = i + 1
Is
validity ≥ requiredDays
AND
totalCost ≤ budget?
Yes
No
Is
totalCost < minCost?
No
Yes
Update
minCost = totalCost
bestPlan = i
Is
bestPlan ≠ -1?
Yes
No
Output
Best Plan: Plan bestPlan
Minimum Cost: ₹ minCost
Output
No eligible plan found
within budget.
Stop

PSEUDOCODE:

```
BEGIN
        INPUT n
        INPUT dataUsed
        INPUT budget

        minCost = INFINITY
        bestPlan = -1

        FOR i = 1 TO n
        INPUT price, dataLimit
        INPUT validity, requiredDays
        INPUT extraRate

        IF dataUsed > dataLimit THEN
        extraCost = (dataUsed - dataLimit) * extraRate
        ELSE
        extraCost = 0
        ENDIF

        totalCost = price + extraCost
        DISPLAY totalCost

        IF validity >= requiredDays AND
        totalCost <= budget AND
        totalCost < minCost THEN
        minCost = totalCost
          bestPlan = i
        ENDIF
        ENDFOR
```

IF bestPlan != -1 THEN

DISPLAY bestPlan, minCost

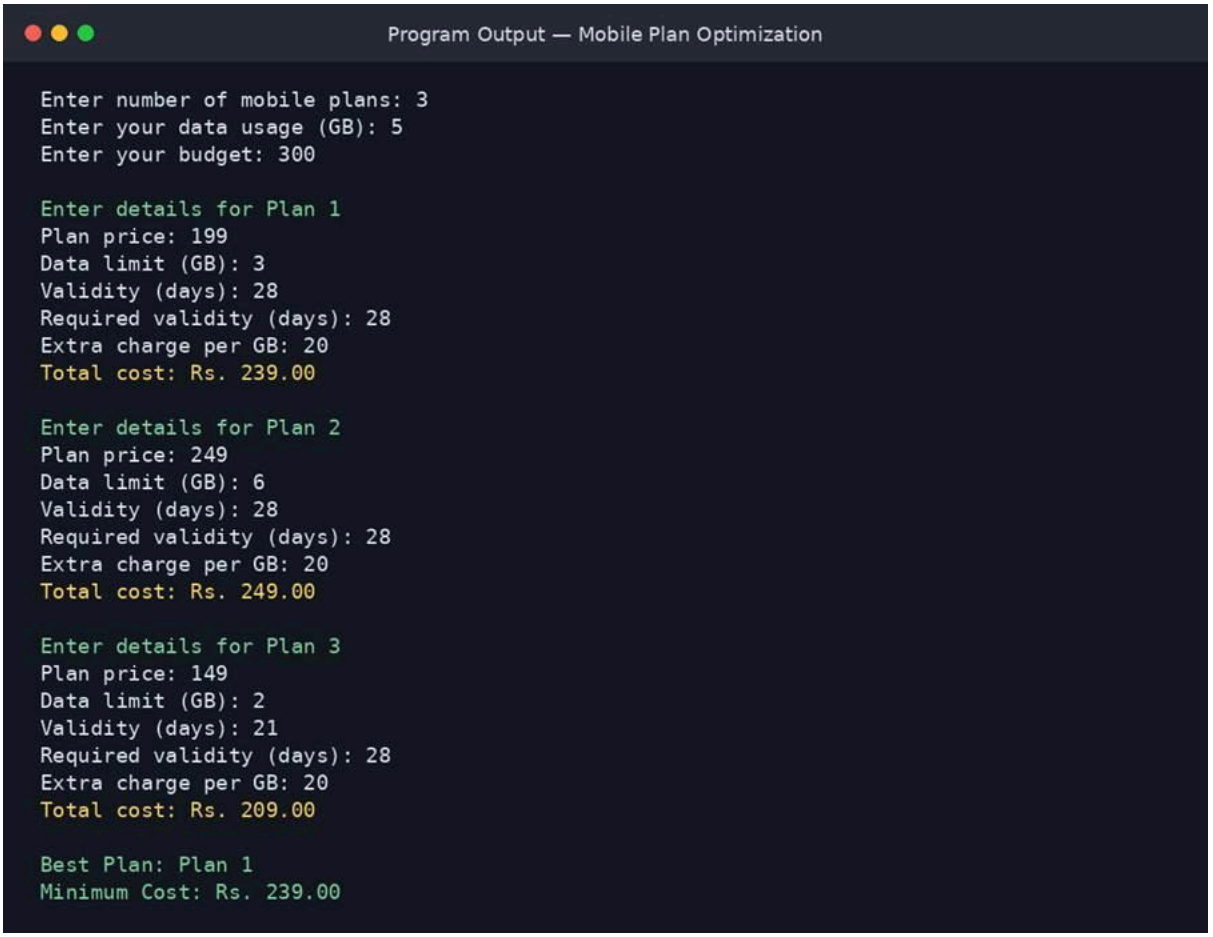
ELSE

DISPLAY "No eligible plan found"

ENDIF

END

SCREENSHOT



OUTPUT:

Example input: 3 plans, data usage 5 GB, budget ₹300, required validity 28 days.

Enter number of mobile plans: 3

Enter your data usage (GB): 5

Enter your budget: 300

Enter details for Plan 1

Plan price: 199

Data limit (GB): 3

Validity (days): 28

Required validity (days): 28

Extra charge per GB: 20

Total cost: Rs. 239.00

Enter details for Plan 2

Plan price: 249

Data limit (GB): 6

Validity (days): 28

Required validity (days): 28

Extra charge per GB: 20

Total cost: Rs. 249.00

Enter details for Plan 3

Plan price: 149

Data limit (GB): 2

Validity (days): 21

Required validity (days): 28

Extra charge per GB: 20

Total cost: Rs. 209.00

Best Plan: Plan 1

Minimum Cost: Rs. 239.00